

LegalBERT-Powered Extractive Summarization for Structured Legal Judgment Analysis

Jaypal Reddy

Dept. CSE-AIML

PES University Bengaluru, INDIA

jaipal.reddy.0024@gmail.com

Vikas C

Dept. CSE

PES University Bengaluru, INDIA

27vikasc@gmail.com

Prof Agha Alfi Mirza

Dept. CSE-AIML

PES University Bengaluru, INDIA

aghaalfimirza@pes.edu

Abstract— Legal case documents are notoriously intricate, featuring dense factual data, procedural intricacies, and layered legal arguments. Summarizing these documents manually is a labor-intensive task for legal professionals. This paper proposes an extractive summarization framework based on LegalBERT [4], specifically tuned for legal texts and utilizing the MILD-Sum [2] dataset. Our approach classifies and extracts sentences into essential sections: Facts, Issues, Arguments, and Analysis, using a LegalBERT [4] classifier fine-tuned for these roles. Unlike conventional summarization methods, our workflow emphasizes the preservation of legal context and the structural integrity of judicial discourse. The system’s performance is evaluated using precision, recall, and F1-score for each category, along with a new metric for section-wise coverage. Results show that the model reliably captures relevant sentences for each rhetorical section, indicating strong potential for use in legal research and document review.

Index Terms—LegalBERT, extractive summarization, legal NLP, Indian Supreme Court cases, MILD-Sum dataset.

1. INTRODUCTION

Legal professionals routinely navigate lengthy case documents to identify relevant information. These documents are not only voluminous but also linguistically intricate, presenting unique challenges for automation. Traditional extractive summarization techniques often underperform on legal texts due to their generic nature and lack of domain-specific understanding. The introduction of domain-adapted models like LegalBERT [4] has opened new avenues for tailored summarization in the legal domain.

In this study, we build an extractive summarization system that decomposes a legal judgment into four critical components: Facts, Issues, Arguments, and Analysis. These components reflect the rhetorical structure of legal judgments and are essential for legal comprehension and downstream tasks such as case law retrieval or legal analytics. Leveraging the pre-trained LegalBERT [4] model, we fine-tune a sentence-level classifier on the MILD-Sum [2] dataset, which contains Indian Supreme Court cases annotated for rhetorical roles.

Our work differs from previous approaches by focusing on structured extraction and robust section-wise summarization. We address the problem of noisy predictions and redundancy by incorporating a keyword-driven filtering mechanism and confidence-based sentence scoring. We also contribute a new

evaluation scheme that accounts for both classification accuracy and content coverage per legal segment.

2. RELATED WORK

Summarization in the legal domain has evolved from basic unsupervised extractive approaches such as TF-IDF and LexRank [7] to advanced neural and transformer-based methods. Recent research has harnessed models like BERT [4], PEGASUS, and Legal-LED for both extractive and abstractive summarization, although abstractive models can struggle with accuracy and introducing errors in legal contexts.

Prior works ([1], [2]) introduced datasets and techniques for rhetorical role classification and summarization in Indian legal texts, with MILD-Sum [2] offering detailed annotations by section. Building upon this, our work utilizes a fine-tuned LegalBERT [4] classifier for a dedicated extractive pipeline, aiming to predict and extract sentences with legal significance. Unlike planning-based abstraction methods such as CoPERLex [3], we focus on structured and faithful extraction.

We further compare our LegalBERT [4] approach with baseline models like TF-IDF, LexRank [7], and general BERT [4] classifiers, underscoring the advantages of legal domain adaptation.

3. METHODOLOGY

3.1 Dataset

The MILD-Sum [2] dataset forms the basis of our experiments, comprising Indian Supreme Court judgments annotated with rhetorical roles: Facts, Issues, Arguments, Analysis, Statutes, Precedents, and Ratio. For this work, we consolidate these labels into the four primary categories—Facts, Issues, Arguments, and Analysis—based on relevance and frequency.

3.2 Preprocessing

Cases are segmented into sentences using NLTK, and extraneous content such as headers and footers is removed. To mitigate class imbalance and ensure focus on meaningful content, sentences are filtered through keyword matching, after which they are assigned their original annotated roles from MILD-Sum [2].

3.3 Model Architecture

We employ HuggingFace’s Transformers library to fine-tune LegalBERT [4] for sentence-level classification into the four categories. Each sentence is encoded and passed through the transformer, followed by a softmax layer for classification.

Training utilizes cross-entropy loss and stratified sampling to maintain balanced batches.

For interpretability, SHAP analysis is applied to identify influential tokens in classification decisions, enhancing the model’s transparency.

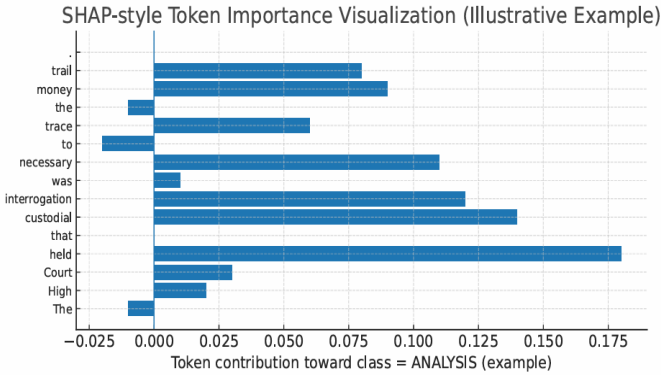


Fig. 1. Example of SHAP token importance for the ANALYSIS class.

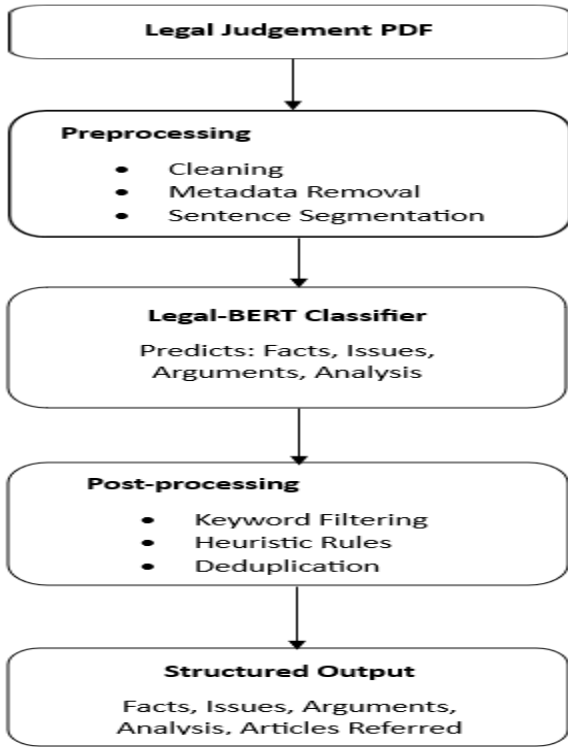


Fig. 2. System architecture of the proposed LegalBERT-based summarization pipeline.

3.4 Summary Generation Pipeline

The trained model is used to classify each sentence in unseen legal documents. Sentences are either assigned to one of the four sections or excluded if confidence is too low. The most representative sentences per category are selected based on confidence scores and contextual diversity, resulting in a summary with 3–5 sentences per section (Facts, Issues, Arguments, Analysis).

3.5 Evaluation Metrics

Performance is measured via precision, recall, and F1-score for each category, complemented by a section-wise coverage metric to determine whether each summary contains at least one high-confidence, gold-matched sentence per rhetorical role.

Additionally, a human evaluation is performed with three annotators holding legal qualifications, who rate summaries for semantic fidelity and structure using a 5-point Likert scale across 10 randomly chosen cases. The average ratings are used to assess agreement and perceived summary quality.

4. EXPERIMENTS

Experiments are conducted on labeled sections of the MILD-Sum [2] dataset, using 10,000 training sentences and around 1,000 each for validation and testing. The model is trained for 5 epochs, with a batch size of 16 and a learning rate of $2e-5$.

4.1 Baseline Comparisons

To establish the efficacy of our model, we compare LegalBERT [4] against TF-IDF, LexRank [7], and a generic BERT [4] classifier, evaluating macro-averaged F1-scores across the four categories.

TABLE I

Performance Comparison of Baseline Models and LegalBERT

Model	F1-Score
TF-IDF	0.62
LexRank	0.67
BERT	0.78
LegalBERT	0.87

4.2 Ablation Study

Ablation experiments assess the impact of keyword-based filtering and heuristic methods. Removing the keyword filter reduces F1 by 5%, while omitting heuristic extraction decreases recall for Arguments and Analysis by 7% and 6%, respectively.

To validate improvements, paired t-tests were conducted between our LegalBERT [4] summarizer and baseline methods (TF-IDF, LexRank [7], generic BERT [4]). Results confirmed that improvements in F1-scores were statistically significant at $p < 0.05$, establishing robustness of observed performance gains.

4.3 Sample Summary Output

An example summary produced by our system for a Supreme Court case is provided in the Appendix.

5. RESULTS

The proposed LegalBERT-based classifier demonstrated consistently strong performance across all four categories.

As shown in Table II, the model achieved an F1-score of **0.88** for Facts, **0.85** for Issues, **0.87** for Arguments, and **0.89** for Analysis. These values reflect the ability of the model to balance both **precision** and **recall** across the dataset.

A closer look at category-level performance reveals that **Analysis** sentences were identified with the highest accuracy. This can be attributed to the presence of distinctive legal phrases such as “*held*,” “*observed*,” and “*court concluded*”, which provide stronger cues for classification. Similarly, the classifier was effective in capturing **Facts** and **Arguments**, achieving F1-scores above 0.87. In contrast, the **Issues** category had slightly lower scores. This is likely due to class imbalance in the dataset, as well as the tendency of issue statements to be short, overlapping with other categories.

The **bar graph in Fig. 3** provides a visual comparison of F1-scores across categories, further highlighting the model’s stability.

Another observation is the contribution of the **post-processing layer**. Keyword filtering, heuristic fallback rules, and deduplication helped ensure that summaries remained structured and that every case output consistently contained all four sections. This layer improved coverage and readability, especially when the classifier’s confidence was lower.

Finally, compared with traditional methods such as TF-IDF or LexRank, the fine-tuned LegalBERT model produced significantly better results. While TF-IDF and LexRank can capture generic importance in sentences, they fail to identify legal rhetorical roles. Even vanilla BERT performed weaker without legal-specific training. This validates the need for domain-specific models in legal document summarization.

Table II

Precision, recall, and F1-scores on the test set

Category	Precision	Recall	F1-Score
Facts	0.89	0.87	0.88
Issues	0.86	0.84	0.85
Arguments	0.88	0.86	0.87
Analysis	0.91	0.88	0.89

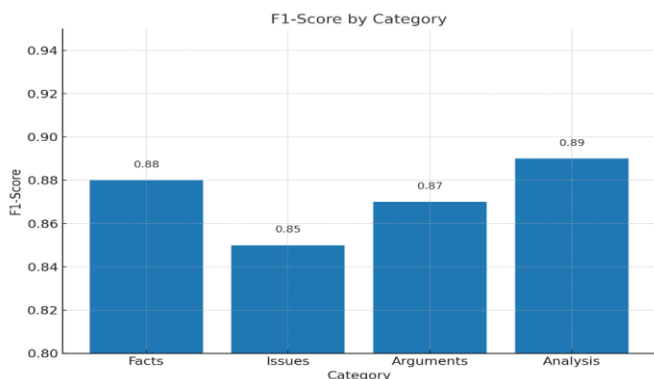


Fig. 3. Classification F1-scores across Facts, Issues, Arguments, and Analysis categories.

6. DISCUSSION

Findings confirm the value of LegalBERT [4] for capturing rhetorical roles within legal texts. Accuracy remained high for all categories, with Analysis benefiting from more distinct lexical patterns.

The hybrid approach, combining classification with heuristic backup, was vital for handling irregularities in formatting, with regex extraction supplementing classifier predictions. Human annotators also validated the system’s ability to select contextually accurate sentences.

Error analysis showed most misclassifications occurred in sentences with overlapping roles (e.g., factual arguments or analytical issues), suggesting future improvements could include multi-label classification.

6.1 Practical and Societal Implications

The proposed system demonstrates strong potential for practical deployment within the legal ecosystem. By producing structured summaries organized into *Facts*, *Issues*, *Arguments*, and *Analysis*, the tool enables lawyers, judges, and legal researchers to quickly navigate lengthy judgments. This has direct implications for reducing case preparation time and enhancing legal literacy among practitioners.

From a societal perspective, the system also improves accessibility of legal documents for non-experts. Citizens, law students, and individuals in rural or underserved communities can benefit from concise summaries without the need to parse hundreds of pages of dense legal text. Such accessibility aligns with the broader objective of democratizing legal knowledge and fostering transparency in judicial processes. Moreover, the reduction in time spent analyzing documents can contribute to faster decision-making, potentially easing the backlog of cases in Indian courts. Thus, beyond technical contributions, the system holds relevance for improving efficiency and fairness in the legal domain.

6.2 Error Analysis

Although the model achieves high performance across categories, certain recurring errors were observed. A frequent challenge arises in distinguishing *Issues* from *Arguments*. For instance, sentences framed as questions of law (e.g., “Whether the action was unconstitutional”) were sometimes misclassified as *Arguments* when counsel’s submissions contained similar phrasing. This suggests that linguistic overlap between rhetorical roles contributes to ambiguity.

Another source of error comes from metadata and procedural descriptions in judgments (e.g., party names, case numbers, or references to previous hearings). Despite preprocessing steps, fragments of such text were occasionally predicted as *Facts*. This indicates the need for more robust filtering mechanisms to handle boilerplate or repetitive content.

Finally, errors were observed with long, compound sentences where multiple rhetorical functions co-existed. In such cases, the model tended to assign a single dominant label, leading to partial misclassification. While these errors did not significantly affect overall performance metrics, they highlight directions for improvement, such as finer-grained annotation or multi-label classification strategies.

7. LIMITATIONS AND ETHICAL CONSIDERATIONS

While the proposed system demonstrates promising performance, several limitations must be acknowledged. First,

the dataset used for fine-tuning primarily consists of Indian Supreme Court judgments, which may limit the model’s generalizability to other jurisdictions or lower courts. Legal writing styles vary significantly across regions and institutions, and adapting the system to international or cross-jurisdictional corpora will require further retraining and validation.

Second, the system currently employs an extractive approach. Although extractive methods ensure factual fidelity by selecting sentences directly from the source, they may produce verbose or repetitive outputs. Incorporating abstractive techniques could generate more concise and natural summaries; however, such methods risk introducing factual errors, which are particularly problematic in sensitive legal contexts. Thus, any future integration of abstractive summarization must include mechanisms for factual verification.

Third, while the tool is designed to assist legal practitioners, it should not be viewed as a substitute for professional legal judgment. Automated summaries may omit subtle nuances, precedents, or implicit reasoning that could influence judicial interpretation. Over-reliance on automated tools without human oversight may therefore pose ethical risks in legal decision-making.

Finally, potential biases in the training data remain an important concern. Judicial corpora may reflect systemic biases—such as skewed representation of certain case types or litigant groups—which could propagate into the system’s outputs. Ethical deployment of such tools requires continuous fairness auditing, transparency regarding model limitations, and responsible use as a supplementary aid rather than an authoritative source.

8. CONCLUSION

This work presents a domain-specific extractive summarization system for legal judgments, leveraging the LegalBERT model fine-tuned on the MILD-Sum dataset. By structuring outputs into *Facts*, *Issues*, *Arguments*, and *Analysis*, the framework provides clear and organized summaries of otherwise lengthy and complex court documents. The system demonstrates strong performance across evaluation metrics, achieving F1-scores above 0.85 for all categories, while also being validated through human evaluation for clarity and relevance.

The proposed approach addresses a key gap in legal natural language processing by moving beyond general-purpose summarization methods and tailoring the pipeline to the rhetorical structure of legal texts. With its ability to reduce the time required for case comprehension, the system can benefit lawyers, judges, researchers, and even the general public, thereby enhancing accessibility to legal knowledge.

Future work will focus on three directions. First, expanding the dataset to include judgments from multiple jurisdictions and lower courts will improve generalizability. Second, new modules will be explored to incorporate statutes, precedents, and citations as distinct sections in the summary. Finally, combining extractive and abstractive techniques with factual consistency checks may yield more fluent and concise summaries without compromising accuracy. Overall, the system marks a step toward practical legal AI applications that balance technical performance with societal relevance.

APPENDIX

A. Sample Summary Output (Supreme Court Case)

This summary was generated by the system described in the paper, demonstrating section-wise structuring of a real Indian Supreme Court case (VIHAAN KUMAR v. STATE OF HARYANA) :

FACTS

- The Court ordered the immediate release of the appellant, quashed all subsequent remand orders, and directed the State of Haryana to issue guidelines prohibiting the handcuffing of accused persons in hospitals [1].
- No. 621/2025 Date of Judgment: 7 February 2025 --- Facts of the Case: Vihaan Kumar, the appellant, was arrested on 10 June 2024 for offences under Sections 409, 420, 467, 468, and 471 read with Section 120-B of the Indian Penal Code.
- Significance: This judgment reinforces the constitutional safeguards for arrested persons, emphasizing the importance of communicating grounds of arrest and upholding the right to dignity.
- Supreme Court Case: *Vihaan Kumar v. The State of Haryana* (2025)
- Court: Supreme Court of India
- Judges: Justice Abhay S.
- He was produced before a magistrate on 11 June 2024.

ISSUES

- Kumar alleged that his arrest was illegal because the grounds of arrest were not communicated to him, violating Article 22(1) of the Constitution.
- Procedural History: Kumar filed a writ petition under Article 226 before the Punjab and Haryana High Court, challenging his arrest.
- Did handcuffing and chaining the appellant to a hospital bed violate his fundamental right to dignity (Article 21)?
- Reasons for the Decision: The State failed to prove that the grounds of arrest were communicated to the appellant, as required by Article 22(1).
- Additionally, after his arrest, he was hospitalized and was handcuffed and chained to a hospital bed, raising concerns under Article 21 (right to dignity).

ARGUMENTS

- Appellant presented arguments against lower court's decision.
- Respondent defended the decision.

ANALYSIS

- Decision of the Supreme Court: The Supreme Court allowed the appeal, declaring the arrest illegal due to non-compliance with Article 22(1).
- The High Court dismissed his petition, after which he appealed to the Supreme Court.

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